

THIRD SEMESTER
MID SEMESTER EXAMINATION

M.Tech. (AI)
September, 2025

AI601: ADVANCED SYSTEMS AND NETWORK SECURITY

Time: 1:30 Hours

Max. Marks: 20

Note: Answer **ALL** questions.
 Assume suitable missing data, if any.
 CO# is course outcome(s) related to the question.

1.[a] As per DES algorithm, the given 64-bit key is

01001000 01100101 01101100 01101100
 01101111 01101111 00100001 00100001

Generate the 56-bit key.

[b] Given the PC-2 permutation table below.

Permuted Choice Two (PC-2)

14	17	11	24	1	5	3	28
15	6	21	10	23	19	12	4
26	8	16	7	27	20	13	2
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

Generate Key1 and Key2 for round 1 and round 2 respectively after performing shift and applying PC-2.

[c] Explain the fiestal structure used in DES. **[1+2+2] [CO2]**

2.[a] Using Vignere cipher, convert the following plain text into cipher text.

Plain text: CYBER THEFT

Key: PROMPT

[b] Explain the concept of confusion and diffusion as per Shannon's theory.

[c] Using the brute-force attack, what are the maximum possible keys for a monoalphabetic cipher. Explain with reason. **[2+2+1] [CO1]**

3.[a] Explain the different ways of public key distribution. **[5] [CO3]**

4.[a] Given a 512-bit block of a message. How can we generate a corresponding 8-bit hash code. Explain using the structure of a hash function.

[2] [CO2]

[b] What will the output for ECB and CBC block cipher mode respectively, given the following details:

Plain text: 00101 00101

Block size: 5 bits

Permuted key: 53124

Initial value: 10110

[3] [CO3]

---Best of Luck---